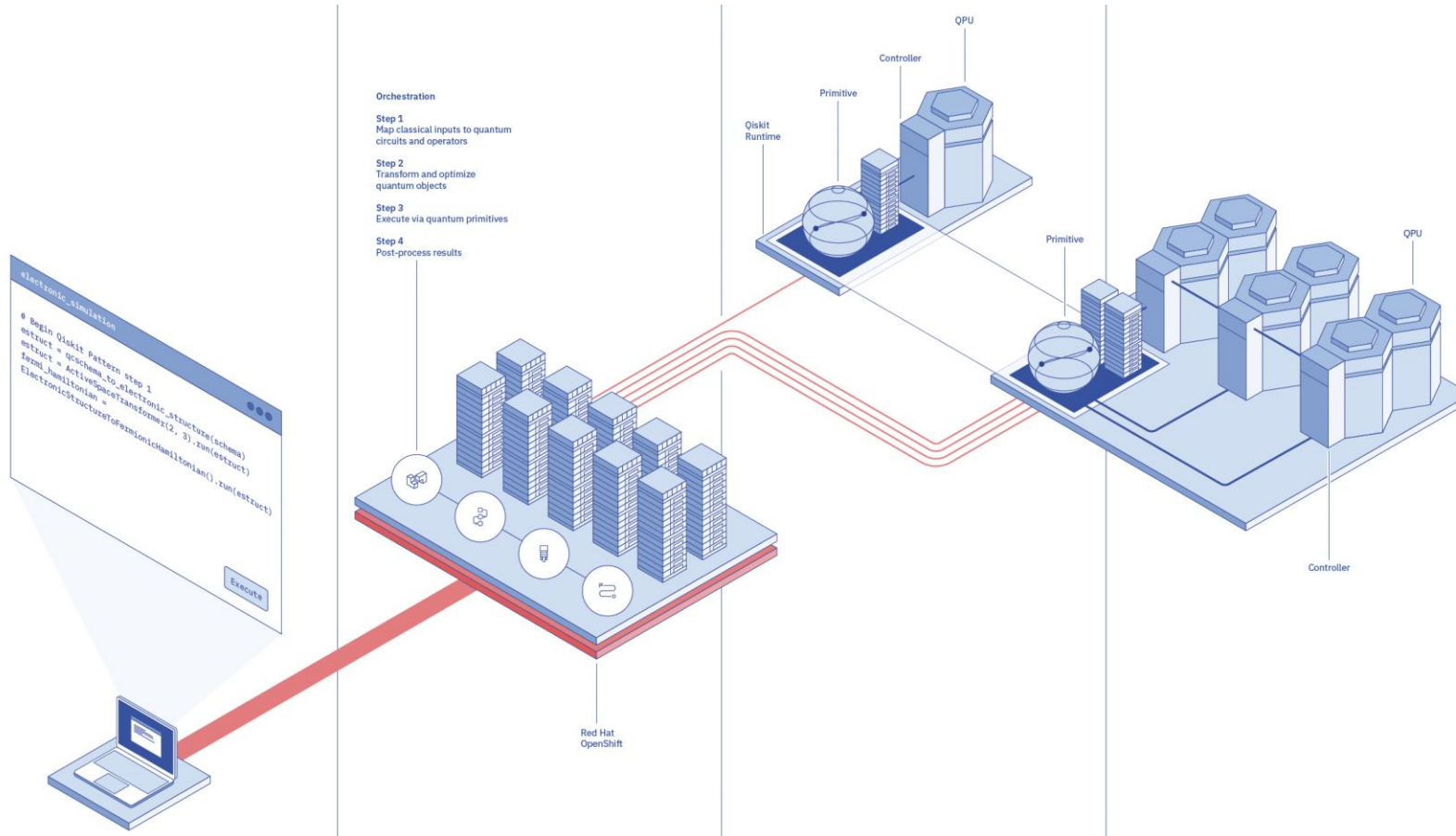


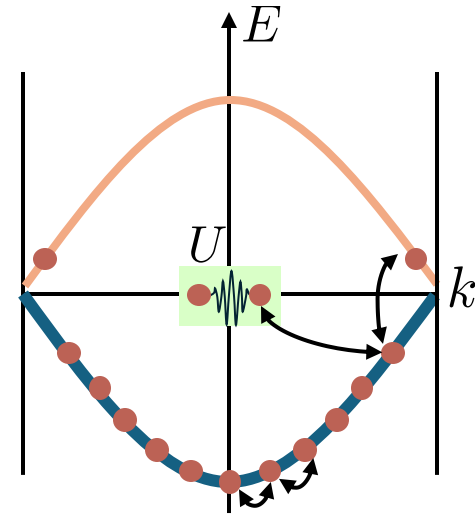
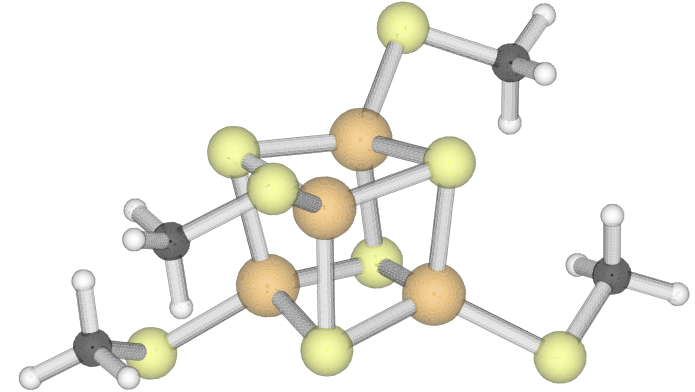
# Sample-based approaches for quantum simulation



Javier Robledo Moreno  
Research Scientist

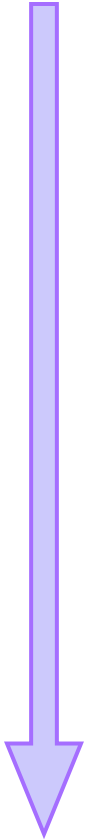
# Quantum Algorithms for ground states

- **Goal:** solving computationally challenging energy-estimation problems in quantum chemistry and materials
- **Applications:** development of new materials, drug discovery, battery design, low-temp. physics



# Algorithmic roadmap

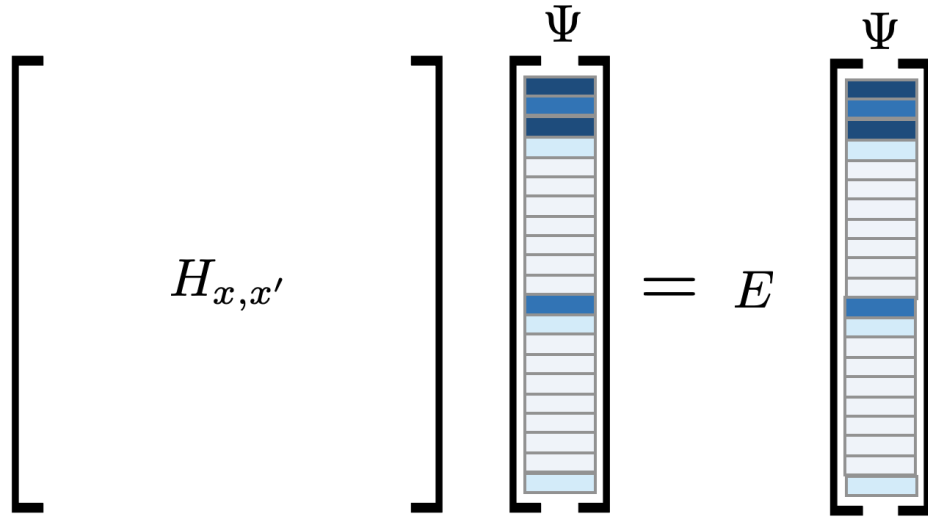
Increasing  
demand on  
quantum  
resources



- Near term:
  - Variational Quantum Eigensolver
  - Sample-Based: SQD | SKQD | SqDRIFT
- Early FT:
  - Quantum Krylov
  - Statistical Phase Estimation
  - Operator Dynamic Mode Decomposition
- FT:
  - Quantum Phase Estimation

Problem statement: sparse eigenstate approximations

# Problem statement



The diagram illustrates the extremal eigenvalue problem. It shows a large square matrix labeled  $H_{x,x'}$  on the left. To its right is a vertical vector labeled  $\Psi$  at the top, which is a column of horizontal bars. Some bars are light blue, some are medium blue, and some are dark blue. To the right of this vector is an equals sign followed by a scalar  $E$ . To the right of  $E$  is another vertical vector, also labeled  $\Psi$  at the top, which is identical in appearance to the first vector.

$$\begin{bmatrix} & \\ & H_{x,x'} \\ & \end{bmatrix} \begin{bmatrix} \Psi \\ \text{vector} \end{bmatrix} = E \begin{bmatrix} \Psi \\ \text{vector} \end{bmatrix}$$

- Extremal eigenvalue problem
- Vector space dimension is exponential
- “Hamiltonian” matrix is exponentially sparse

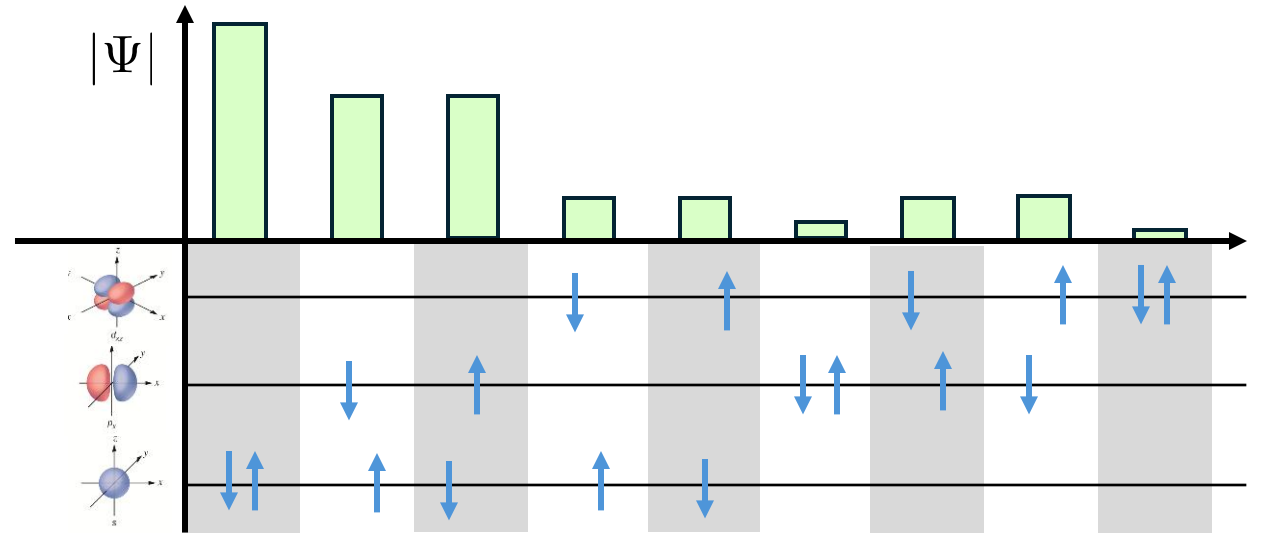
$$I = \sum_{\mathbf{X}} |\mathbf{X}\rangle \langle \mathbf{X}|$$
$$\mathbf{X} \in \{0, 1\}^N$$

# Problem statement

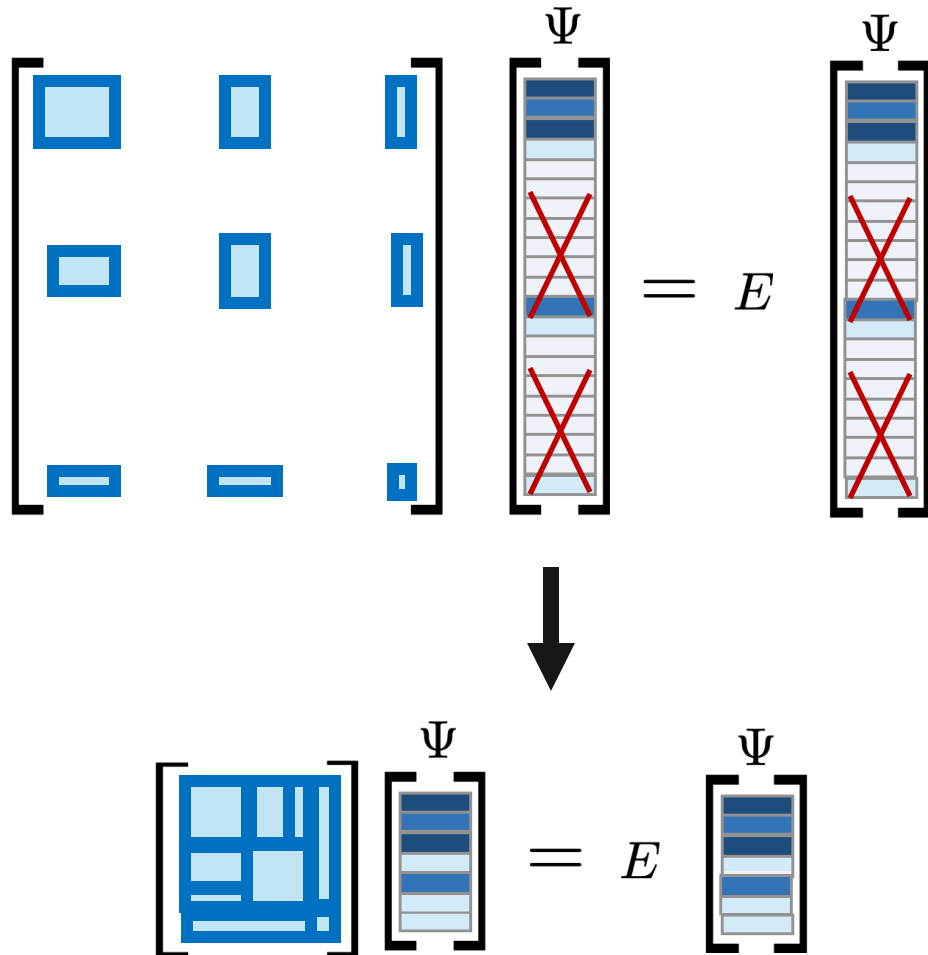
$$H_{x,x'} = E \Psi$$

$$I = \sum_{\mathbf{X}} |\mathbf{X}\rangle \langle \mathbf{X}|$$

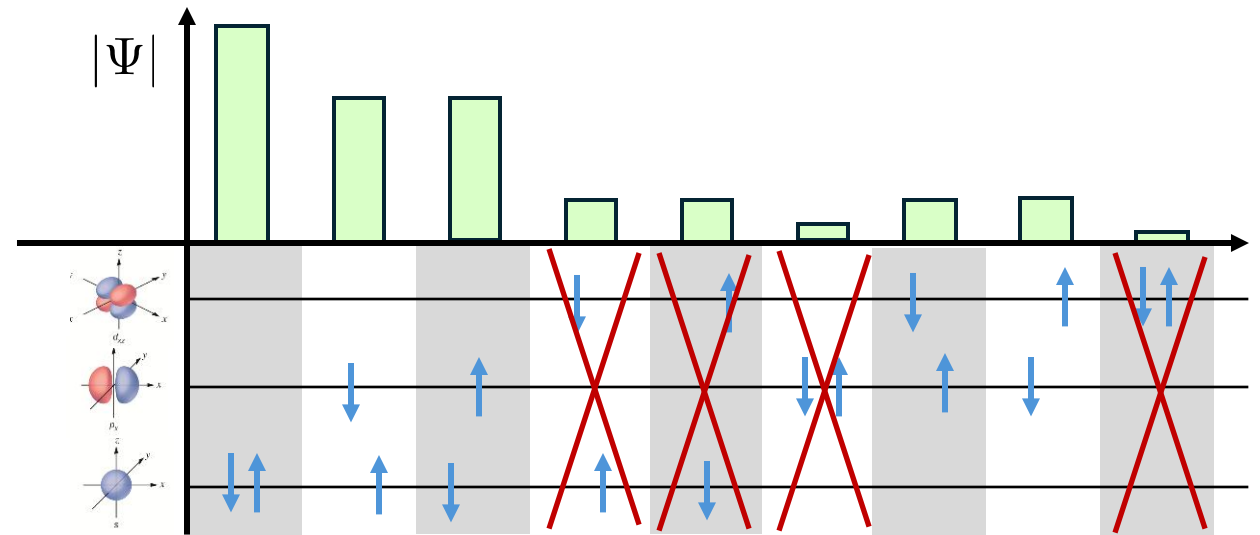
- Extremal eigenvalue problem
- Vector space dimension is exponential
- “Hamiltonian” matrix is exponentially sparse
- The eigenvectors have poly. support (physical intuition)



# Problem statement



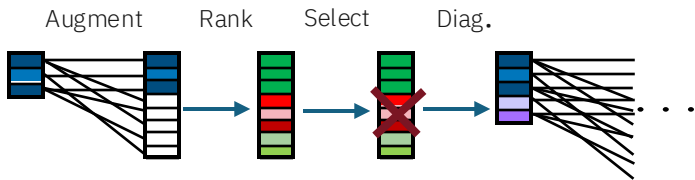
- Thought experiment: we are given the subspace of relevant amplitudes
- We can remove “less relevant”
- Tractable eigenvalue problem



# Searching for relevant dimensions

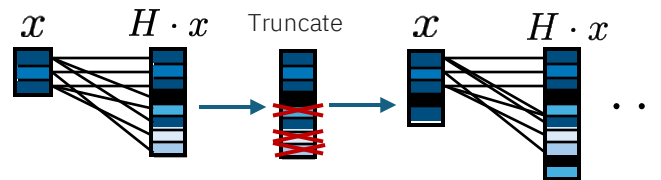
## Selected configuration interaction

- Iterative search (Selected Configuration Interaction)
- Physics intuition + perturbation theory



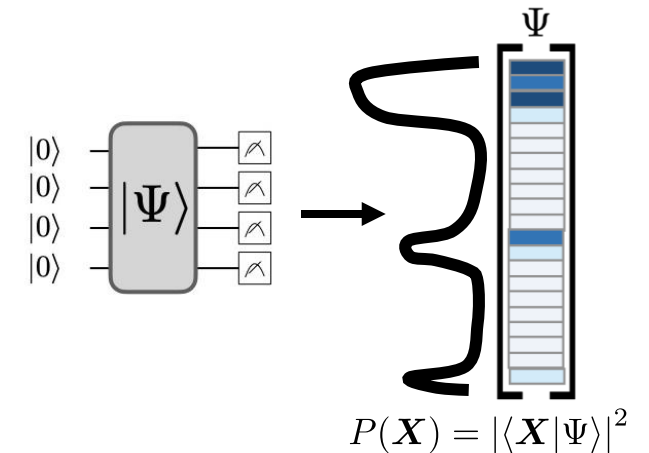
## Truncated power method (s)

- Hamiltonian application followed by truncation
- No subspace diagonalizations at each iteration
- Provable convergence



## SQD

- Quantum circuit produces samples
- Provable convergence



$$\begin{bmatrix} \text{Matrix} \end{bmatrix} \begin{bmatrix} \Psi \end{bmatrix} = E \begin{bmatrix} \Psi \end{bmatrix}$$



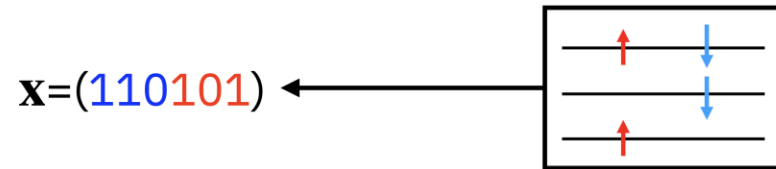
# Sample-based quantum algorithms

# Sample-based quantum diagonalization (SQD)

- Quantum selection: use one or more quantum circuits to sample binary strings

$$P(\mathbf{x}) = |\langle \mathbf{x} | \psi_{QC} \rangle|^2$$

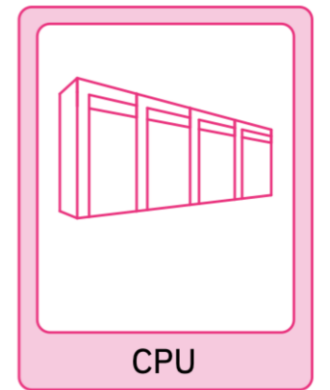
- Binary string: Parametrizes a Slater determinant



- Configuration interaction: approximates eigenstates with sparse CI vectors

$$|\psi\rangle = \sum_{\mathbf{x} \in \chi} c_{\mathbf{x}} |\mathbf{x}\rangle$$

$$\sum_{\mathbf{x} \in \chi} \langle \mathbf{y} | H | \mathbf{x} \rangle c_{\mathbf{x}} = E_0 c_{\mathbf{y}}$$



# How do we choose the quantum circuits?

## Heuristic

- Physically-inspired ansatze (Local Unitary Cluster Jastrow)
- Can either be optimized or initialized from approximate classical electronic structure methods

Science Advances 11 (25), eadu9991 (2025)

SQD

## With convergence guarantees

- Sample from the members of a Krylov basis
- Time-evolution circuits may be hardware friendly and are a Krylov basis
- Not suitable for ab-initio

arXiv:2501.09702

SKQD

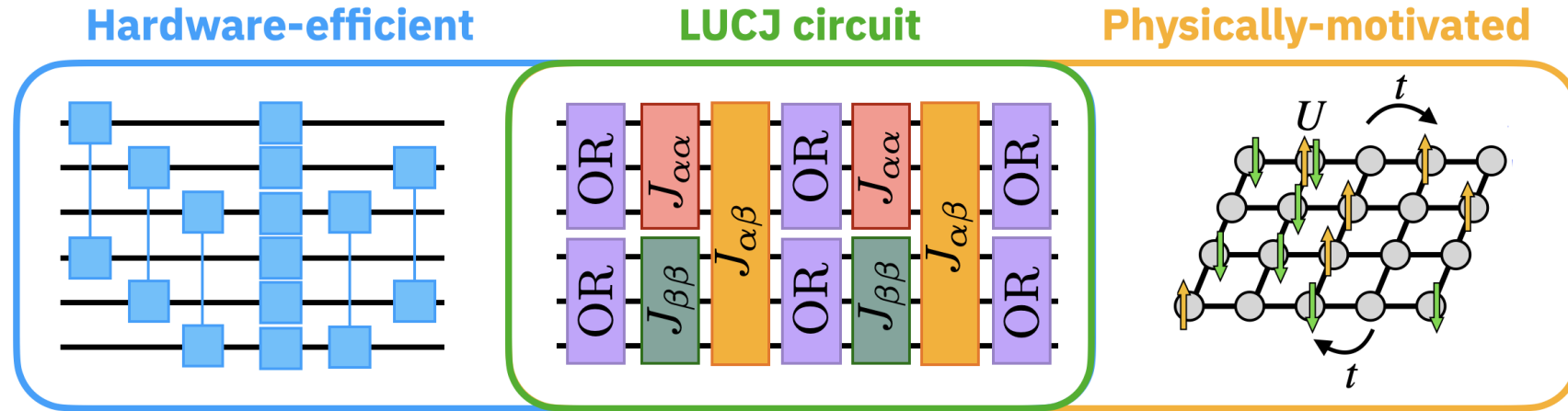
## With convergence guarantees + hardware friendly

- Randomized time evolution a la qDRIFT

arXiv:2508.02578

SqDRIFT

# Local Unitary Cluster Jastrow



Compatible with hardware  
connectivity/gates/depth...  
Hard to initialize and optimize

Parameters from classical  
calculations, easy to optimize  
All-to-all connectivity, high circuit  
depth and gate count



Local Unitary Cluster Jastrow

Physical motivation (derived from coupled-cluster)

Hardware friendliness (device-specific connectivity)

# Local Unitary Cluster Jastrow

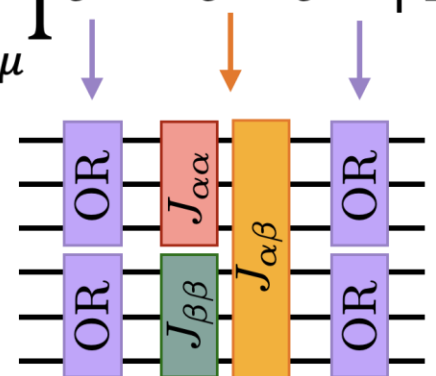
- Unitary coupled-cluster singles and doubles (UCCSD)

$$|\Psi\rangle = e^{T-T^\dagger} |\Phi_0\rangle \quad T = \sum_{ai} t_i^a \hat{c}_a^\dagger \hat{c}_i + \sum_{abij} t_{ij}^{ab} \hat{c}_a^\dagger \hat{c}_b^\dagger \hat{c}_j \hat{c}_i$$

- Jastrow form of the UCCSD:

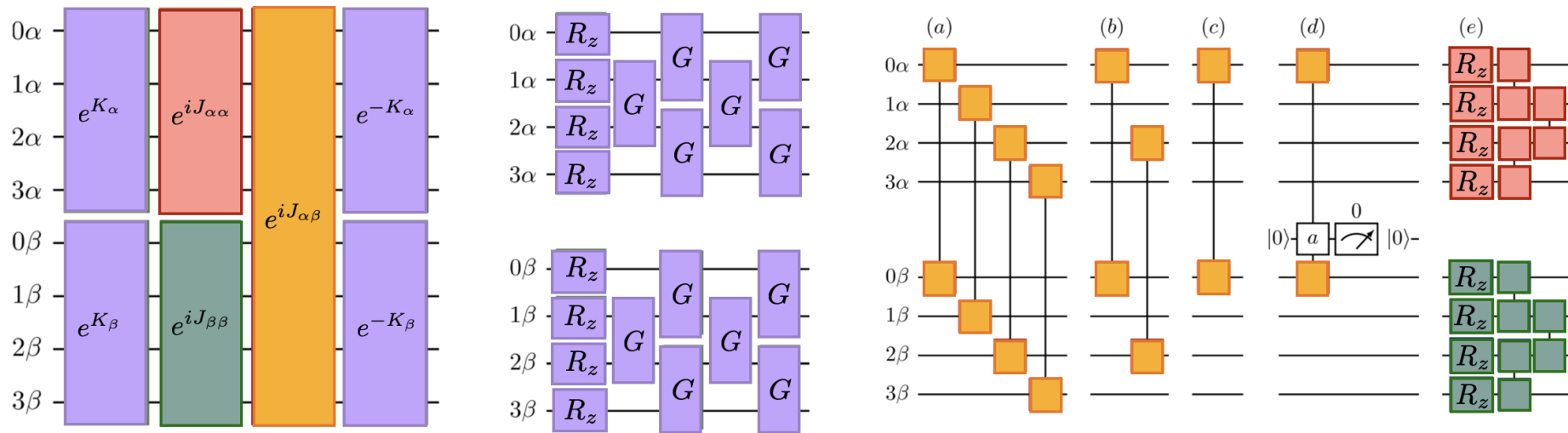
$$T - T^\dagger \simeq \sum_{\mu} e^{-K_{\mu}} iJ_{\mu} e^{K_{\mu}}, \quad K_{\mu} = \sum_{pr,\sigma} K_{pr}^{\mu} \hat{c}_{p\sigma}^\dagger \hat{c}_{r\sigma}, \quad J_{\mu} = \sum_{pr,\sigma\tau} J_{pr}^{\sigma\tau} \hat{n}_{p\sigma} \hat{n}_{r\tau}$$

- Local Unitary Cluster Jastrow ansatz\*

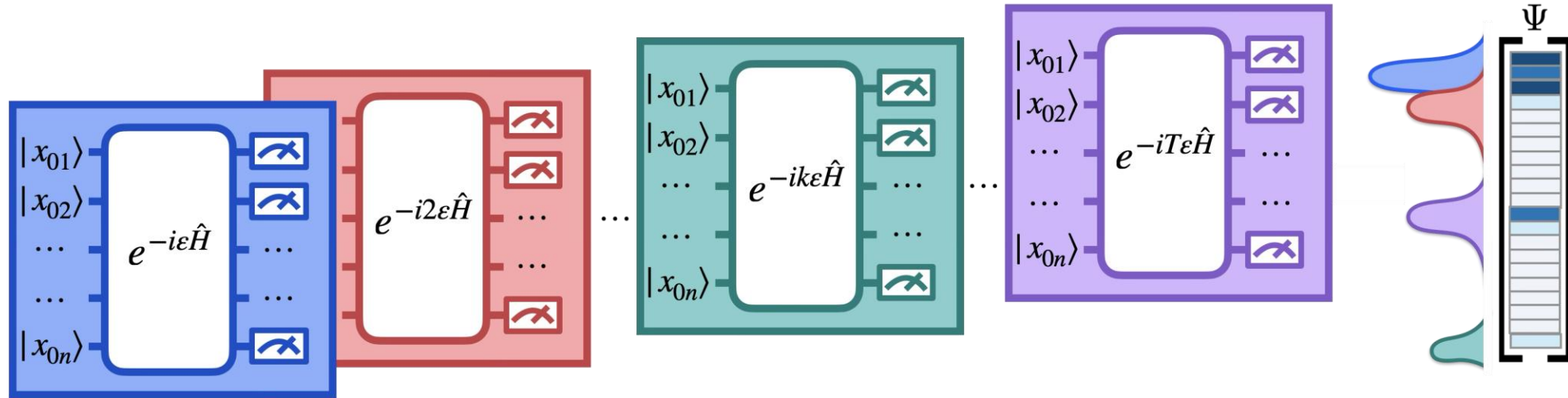
$$|\Psi\rangle \simeq \prod_{\mu} e^{-K_{\mu}} e^{iJ_{\mu}} e^{K_{\mu}} |\Phi_0\rangle$$


# Local Unitary Cluster Jastrow

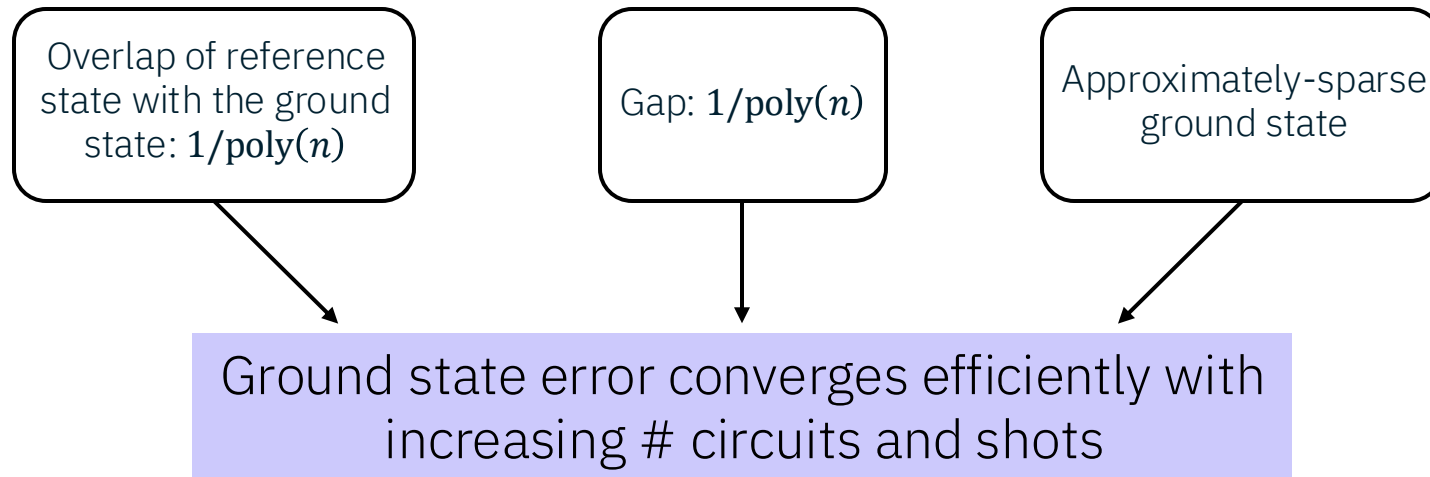
- These circuits can take as input parameters from classical solutions
- There is evidence that they can perform better than their classical counterpart\*
- The circuits can be hard to simulate classically
- Compatible with quantum hardware



# Time-evolution based with provable convergence (SKQD)



- Sample from multiple time-evolved circuits over reference state:



# Time-evolution based with provable convergence (SqDRIFT)

$$H = \sum_{pqrs} h_{pqrs} \hat{c}_p^\dagger \hat{c}_q^\dagger \hat{c}_s \hat{c}_r$$

$$\hat{h}_1 = h_{0101} \hat{c}_0^\dagger \hat{c}_1^\dagger \hat{c}_1 \hat{c}_0$$

$$\hat{h}_2 = h_{0102} \hat{c}_0^\dagger \hat{c}_1^\dagger \hat{c}_2 \hat{c}_0$$

■  
■  
■

Sample terms

$$\mathcal{H}_j = \{\hat{h}_k \mid \hat{h}_k \sim |h_k|\}$$

$$|\Psi_j\rangle$$

=

$$|\phi_{\text{ref}}\rangle$$

$$\prod_{\hat{h}_k \in \mathcal{H}_j} \exp(-i\Delta t \hat{h}_k)$$



# Effect of noise and config. recovery

# Subspace diagonalization and noise tolerance

$$\vec{x} = 0011$$

$$\vec{x} = 1010$$

$$\vec{x} = 1100$$



$$H_{\mathcal{S}^{(k)}} = P_{\mathcal{S}^{(k)}} H P_{\mathcal{S}^{(k)}}$$

$$\text{with } P_{\mathcal{S}^{(k)}} = \sum_{\mathbf{x} \in \mathcal{S}^{(k)}} |\mathbf{x}\rangle \langle \mathbf{x}|$$

$$\begin{pmatrix} \langle \vec{x} | \vec{x} \rangle & \langle \vec{x} | \vec{x} \rangle & \langle \vec{x} | \vec{x} \rangle \\ \langle \vec{x} | \vec{x} \rangle & \langle \vec{x} | \vec{x} \rangle & \langle \vec{x} | \vec{x} \rangle \\ \langle \vec{x} | \vec{x} \rangle & \langle \vec{x} | \vec{x} \rangle & \langle \vec{x} | \vec{x} \rangle \end{pmatrix}$$

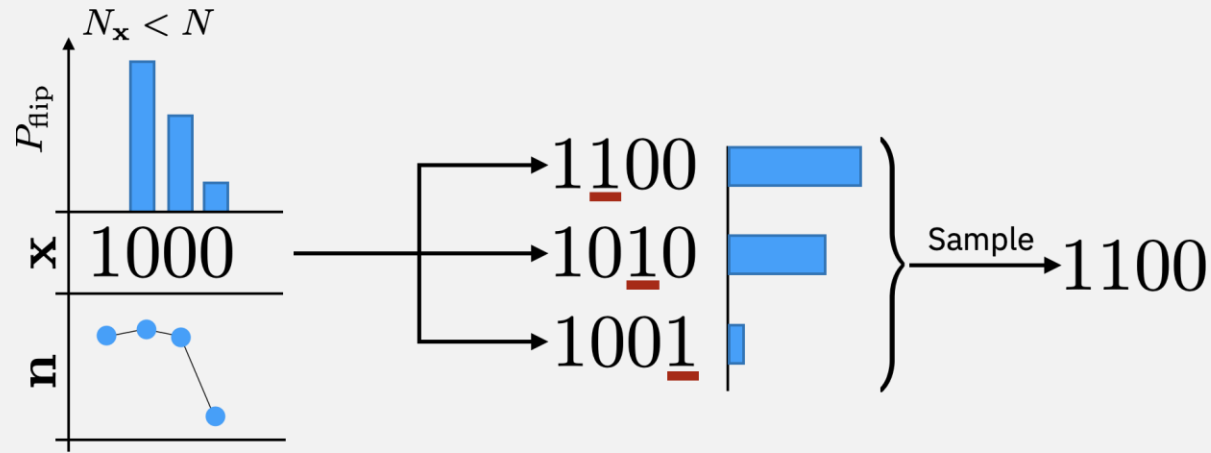


$$H_{\mathcal{S}^{(k)}} \big|^{(k)-} = E^{(k)} \big|^{(k)-}$$

$$\big|^{(k)}i\rangle = \frac{1}{\sqrt{2}} a + 0 \cdot b + \frac{1}{\sqrt{2}} c$$

$$|\text{GroundTruth}\rangle = \frac{|1100\rangle + |0011\rangle}{\sqrt{2}}$$

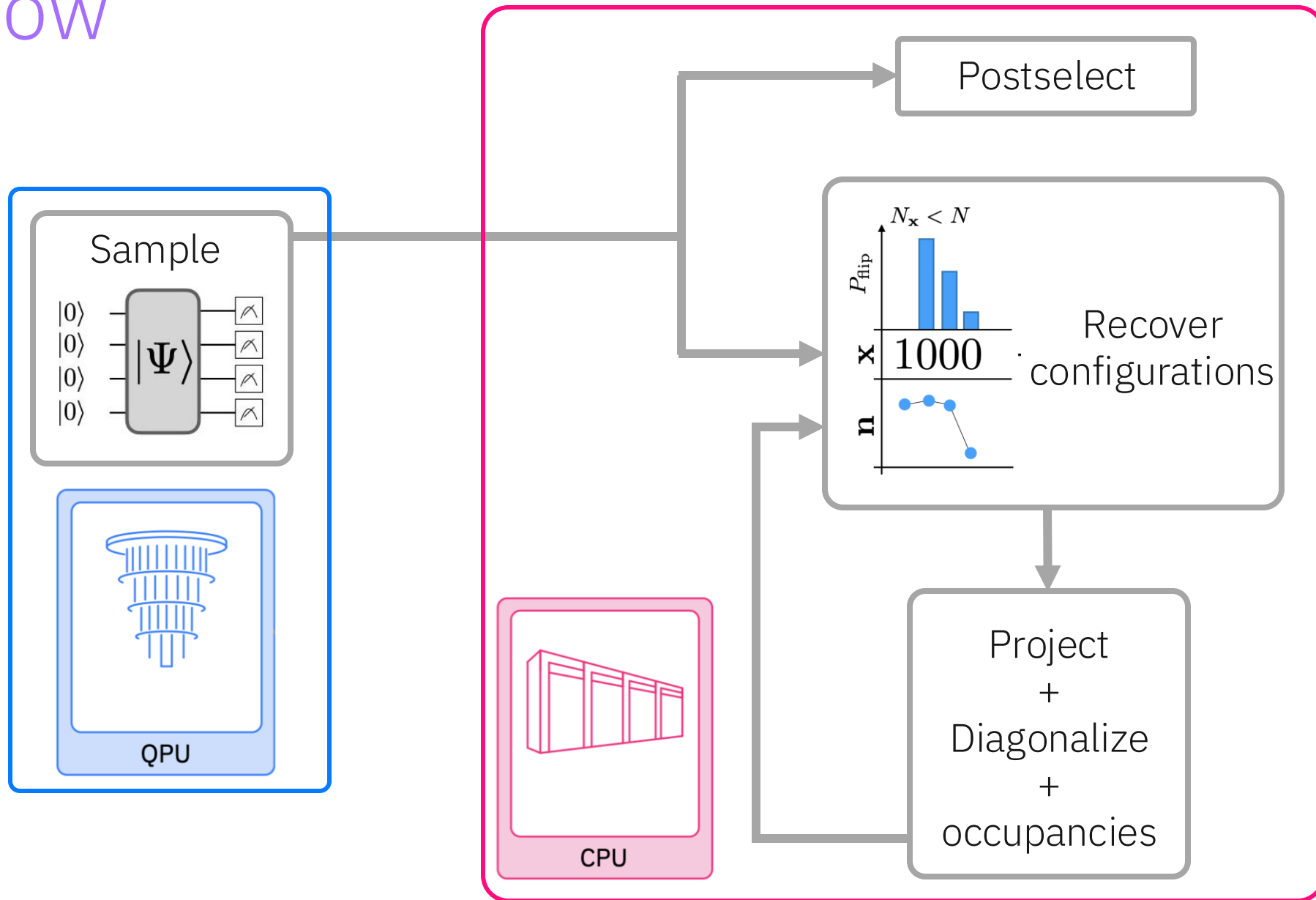
# Self-consistent configuration recovery



$$n_{p\sigma} = \langle \psi | \hat{n}_{p\sigma} | \psi \rangle = \langle \psi | \hat{c}_{p\sigma}^\dagger \hat{c}_{p\sigma} | \psi \rangle$$

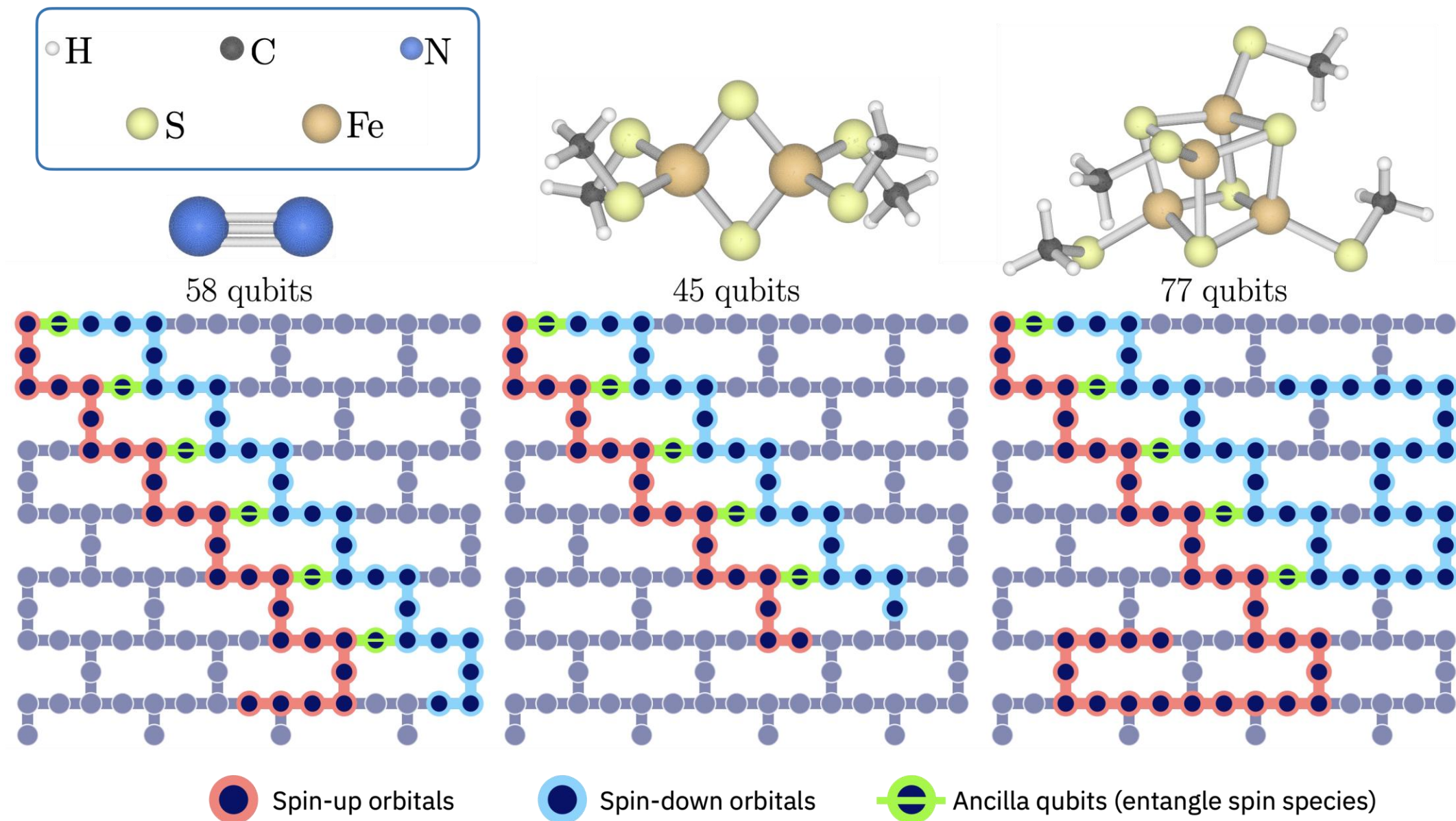
Particle number restored by flipping bits according to a probability informed by the difference between the value of the bit and the average occupancy of the orbital and the filling factor

# Workflow



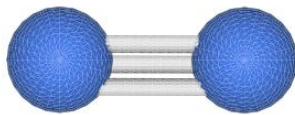
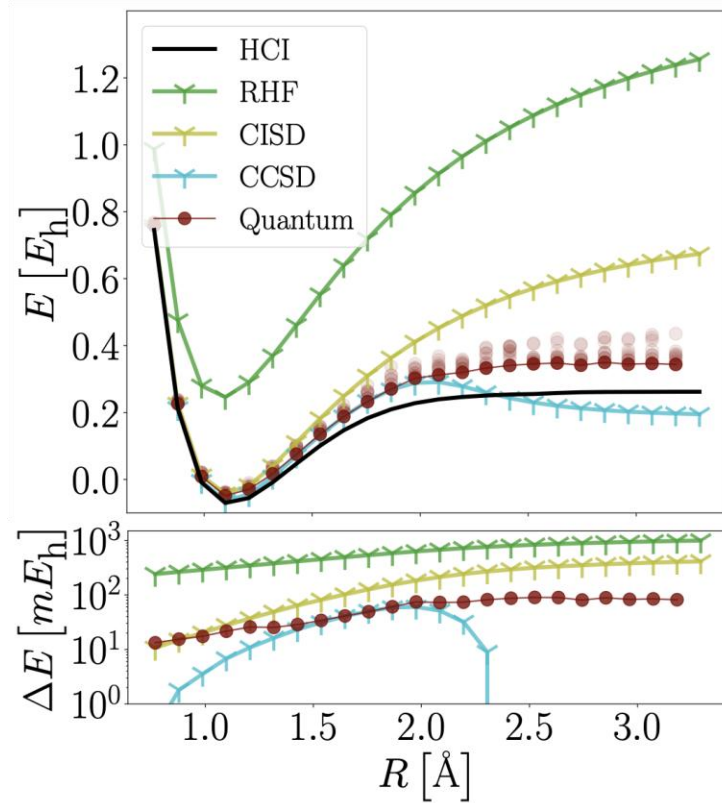
# Applications

# First experimental results: LUCJ circuits

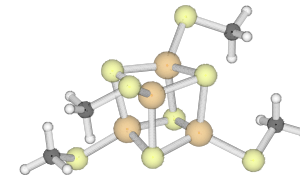
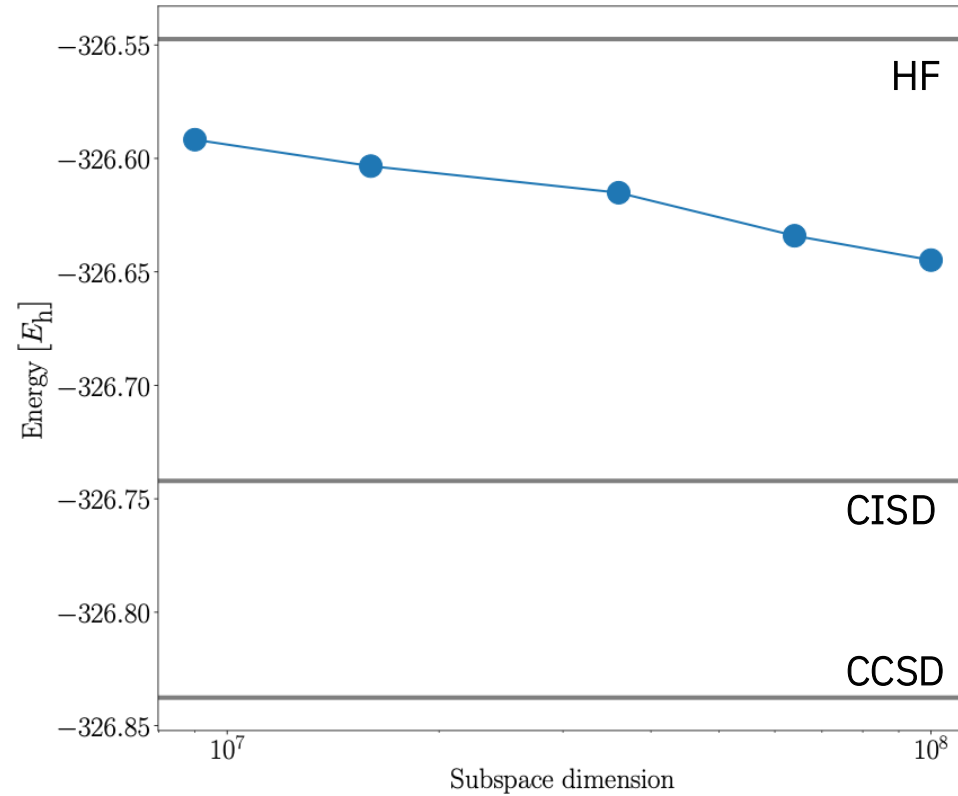


# First experimental results: LUCJ circuits

$N_2$ : Bond breaking on large basis set

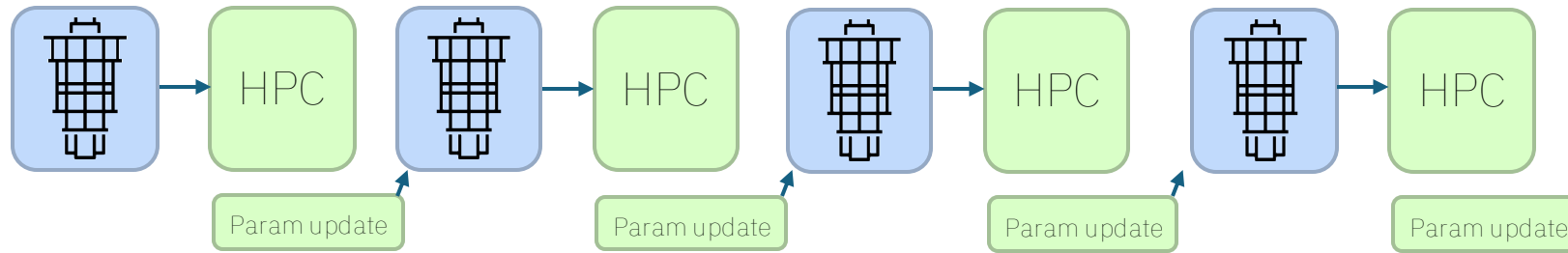


$Fe_4S_4$ : Pushing hardware capabilities

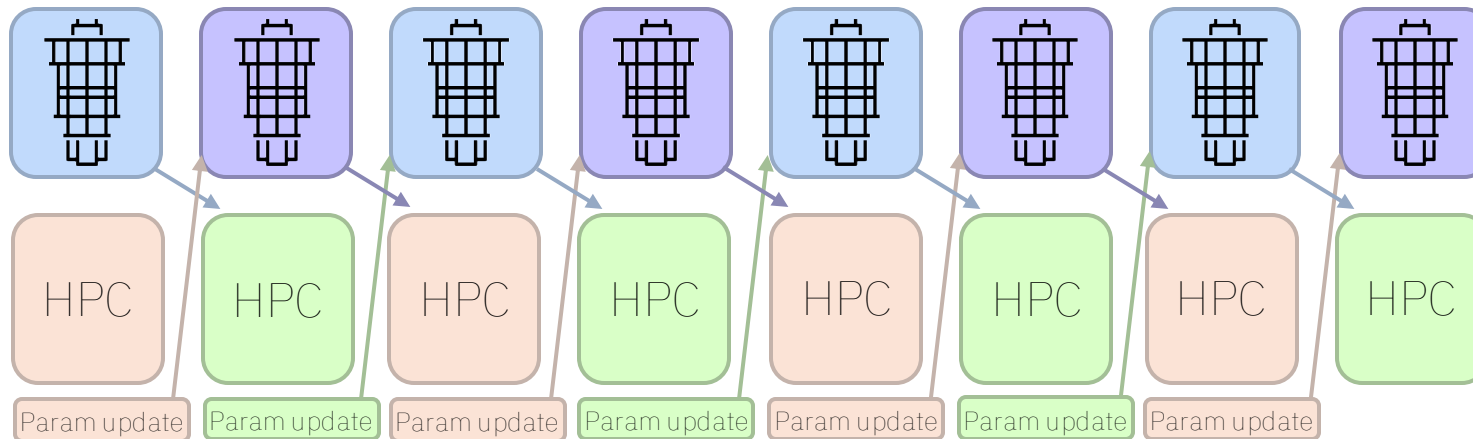


# Parameter optimization

- Quantum computer is tightly coupled with a classical computer to lower energy estimates
  - Iterative solvers lead to idle times



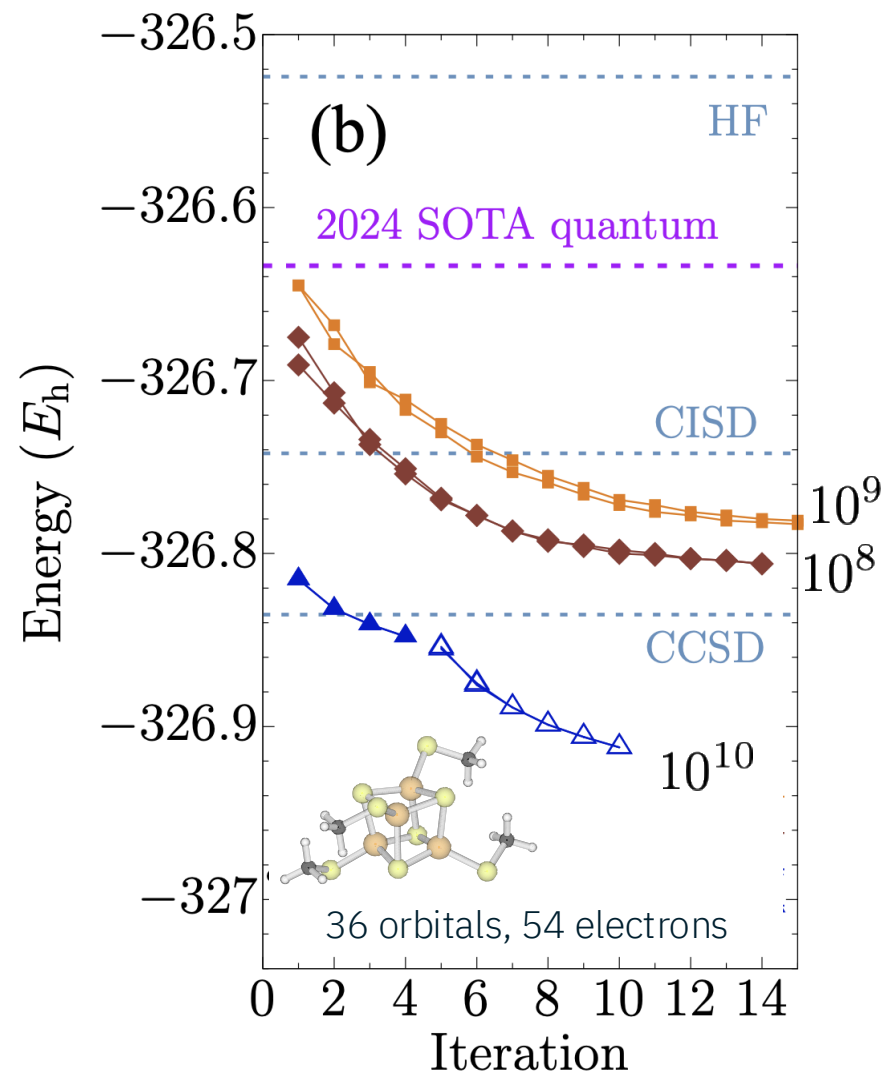
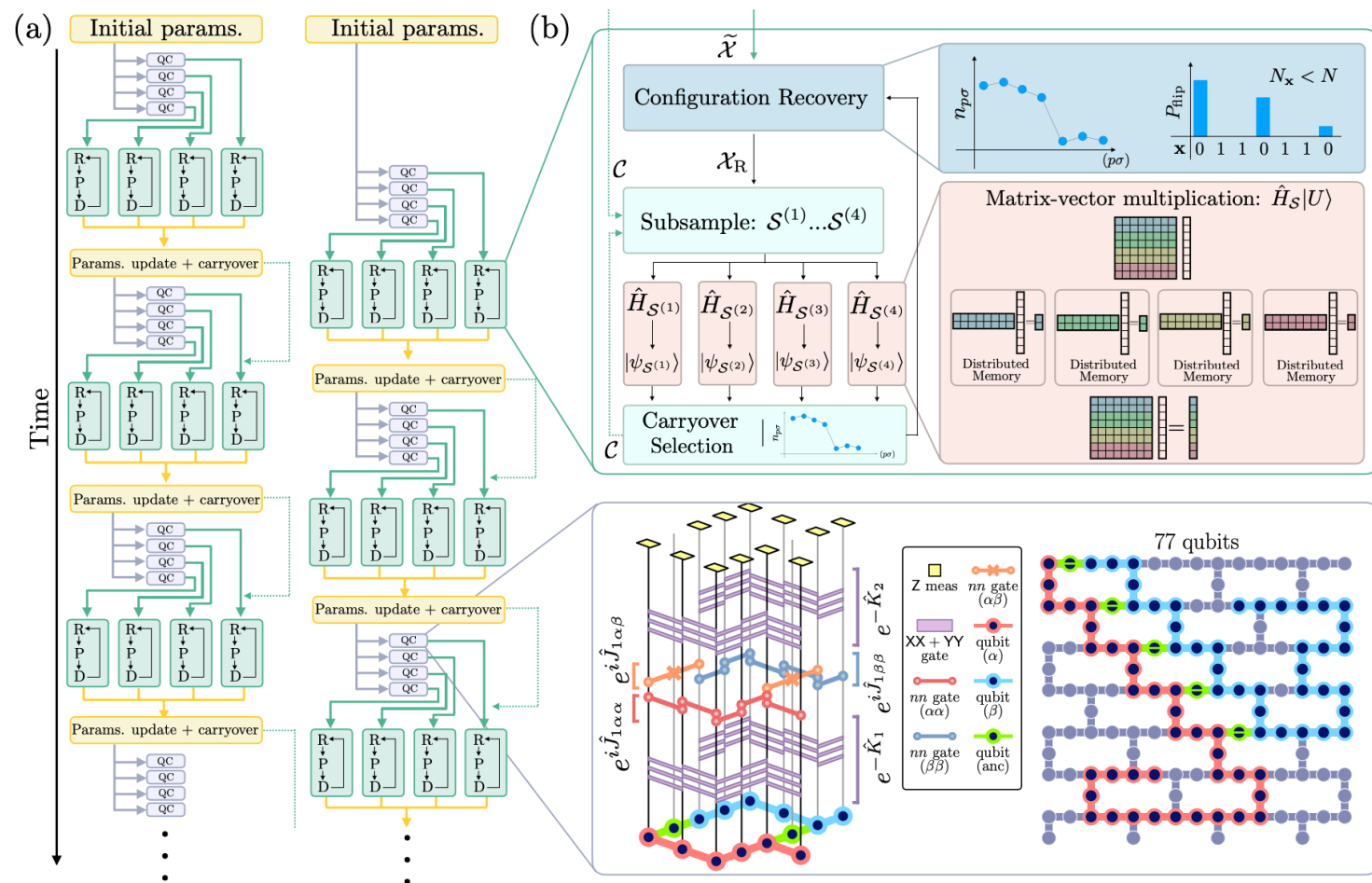
- Differential evolution well suited for optimal scheduling





# Large-scale SQD circuit optimization

Differential-evolution optimization with two populations: no idle time



# Extended SQD: excited States

- Motivation: Quantum subspace expansion

$$|\psi_n\rangle = \sum_i c_i^n |e_i\rangle \longrightarrow |e_i\rangle = \hat{e}_i^\dagger |\psi_0\rangle$$

$$\text{For instance } \{\hat{e}_i^\dagger\} = \{c_i^\dagger c_j, c_i^\dagger c_j^\dagger c_k c_l\}$$

$$\begin{bmatrix} \langle e_j | H | e_i \rangle \end{bmatrix} \begin{bmatrix} c_j^n \end{bmatrix} = E_n \begin{bmatrix} \langle e_j | e_i \rangle \end{bmatrix} \begin{bmatrix} c_j^n \end{bmatrix}$$

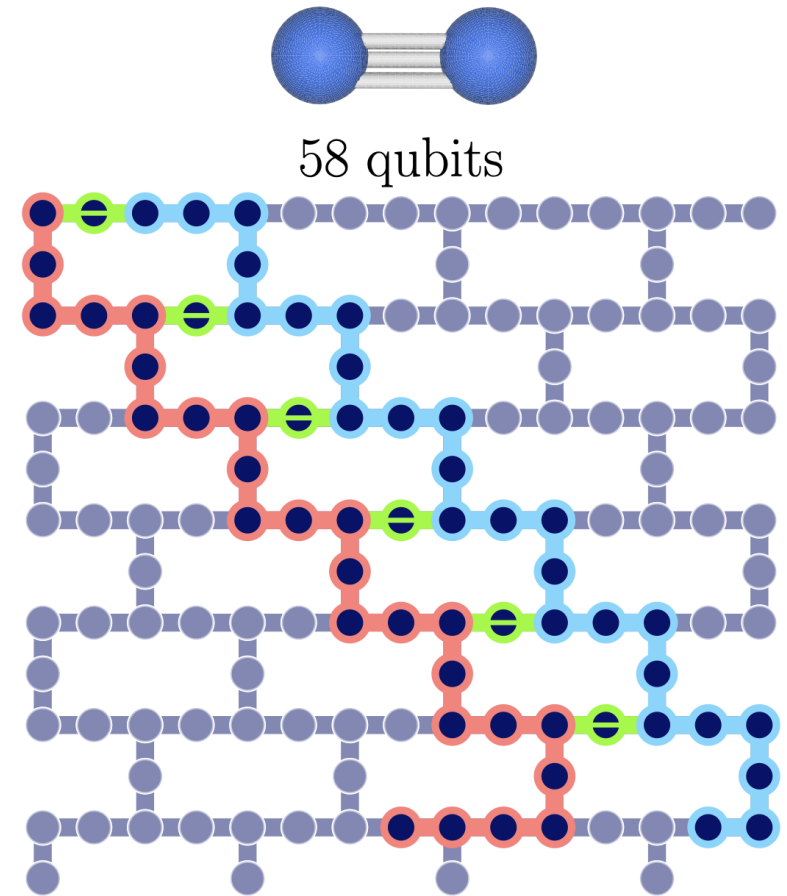
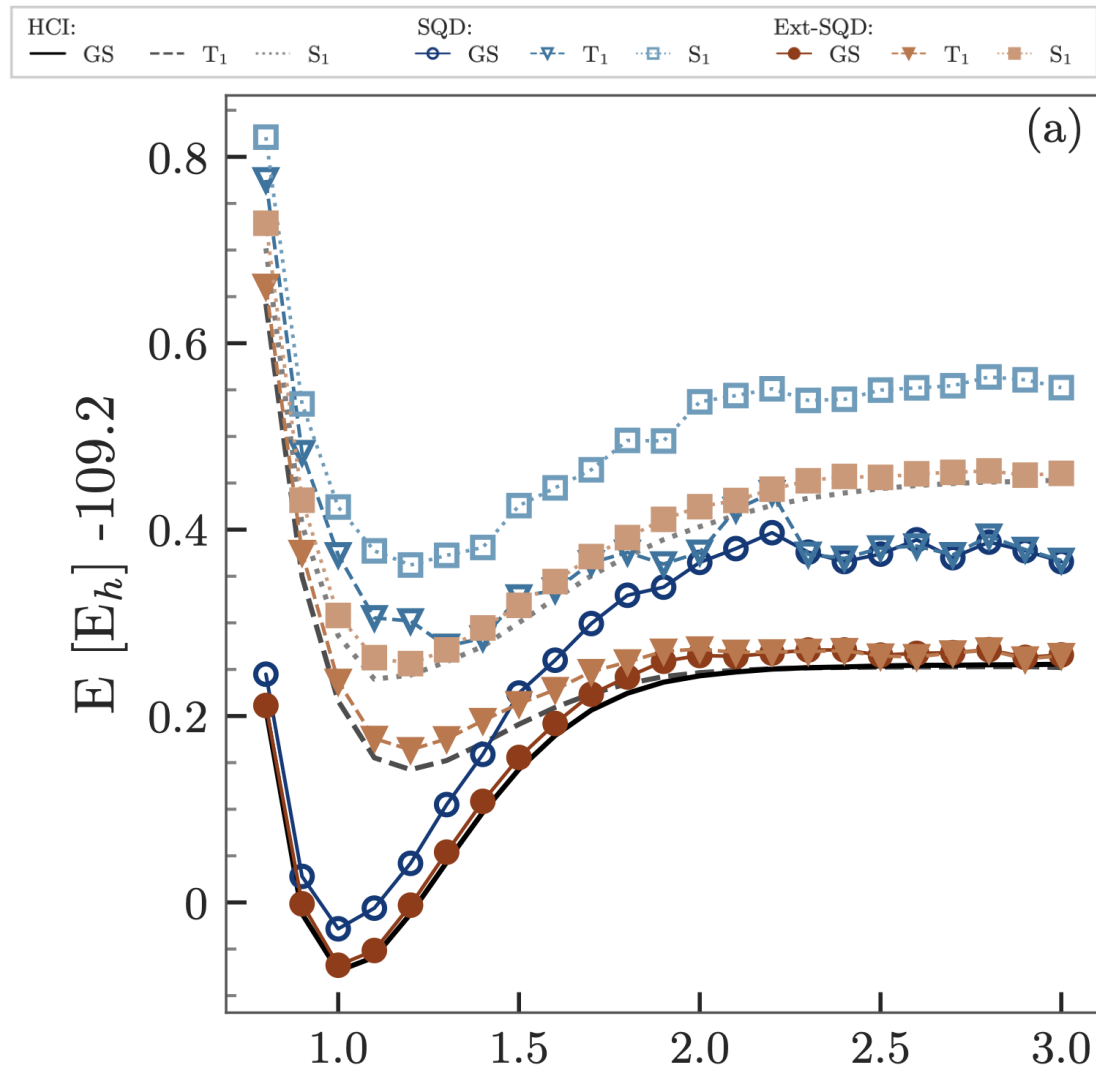
- Sample-based case:

$$|\psi_0\rangle = \sum_{\mathbf{x} \in \chi} c_{\mathbf{x}} |\mathbf{x}\rangle \longrightarrow |e_i\rangle = \sum_{\mathbf{x} \in \chi} c_{\mathbf{x}} \hat{e}_i^\dagger |\mathbf{x}\rangle \longrightarrow |e_i\rangle = \sum_{\mathbf{x} \in \chi} c_{\mathbf{x}} |e_i^\dagger(\mathbf{x})\rangle$$

$$\sum_{\mathbf{x} \in \epsilon} \langle \mathbf{y} | H | \mathbf{x} \rangle c_{\mathbf{x}}^n = E_n c_{\mathbf{y}}^n$$

$$\epsilon = \left\{ e_i^\dagger(\mathbf{x}) \mid \mathbf{x} \in \chi \text{ and for all } e_i^\dagger \right\}$$

# Extended SQD: excited States



# Lattice models: SKQD

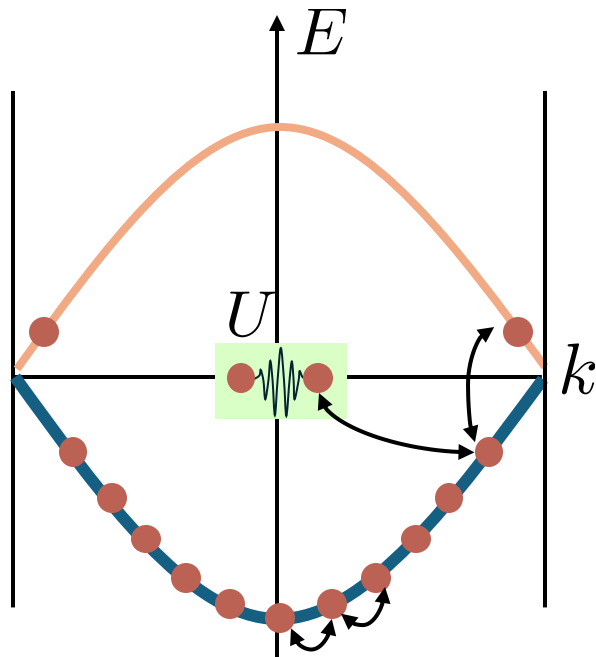
Free electrons coupled to interacting impurities

$$H = H_{\text{bath}} + H_{\text{imp.}} + H_{\text{hyb.}}$$

$$H_{\text{bath}} = \sum_{k\sigma} \varepsilon_k \hat{c}_{k\sigma}^\dagger \hat{c}_{k\sigma}$$

$$H_{\text{hyb.}} = \sum_{k\sigma} V_k \left( \hat{c}_{k\sigma}^\dagger \hat{d}_\sigma + \hat{d}_\sigma^\dagger \hat{c}_{k\sigma} \right)$$

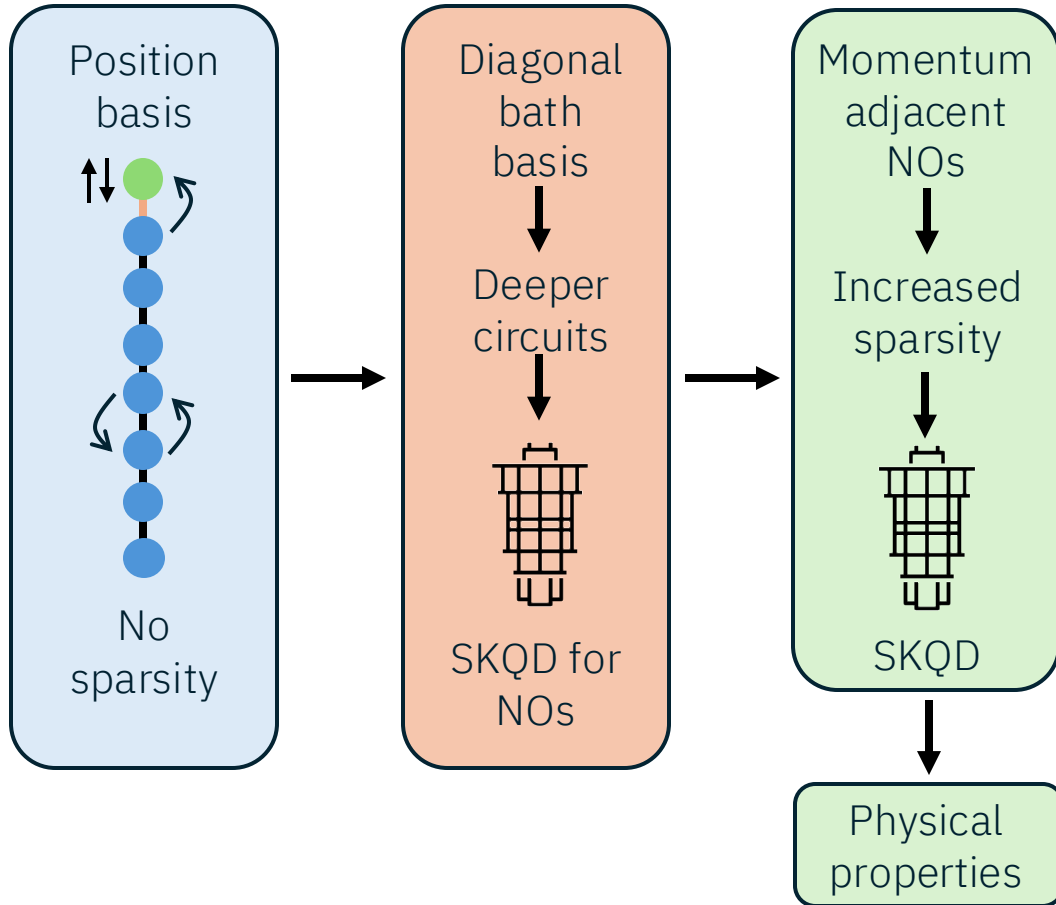
$$H_{\text{imp.}} = U \hat{d}_\uparrow^\dagger \hat{d}_\uparrow \hat{d}_\downarrow^\dagger \hat{d}_\downarrow$$



Relevant to **simplify real materials calculations**  
on DMFT workflows

Extensively used to **benchmark numerical techniques**

# Sparsity is basis dependent



$$H = H_{\text{bath}} + H_{\text{imp.}} + H_{\text{hyb.}}$$

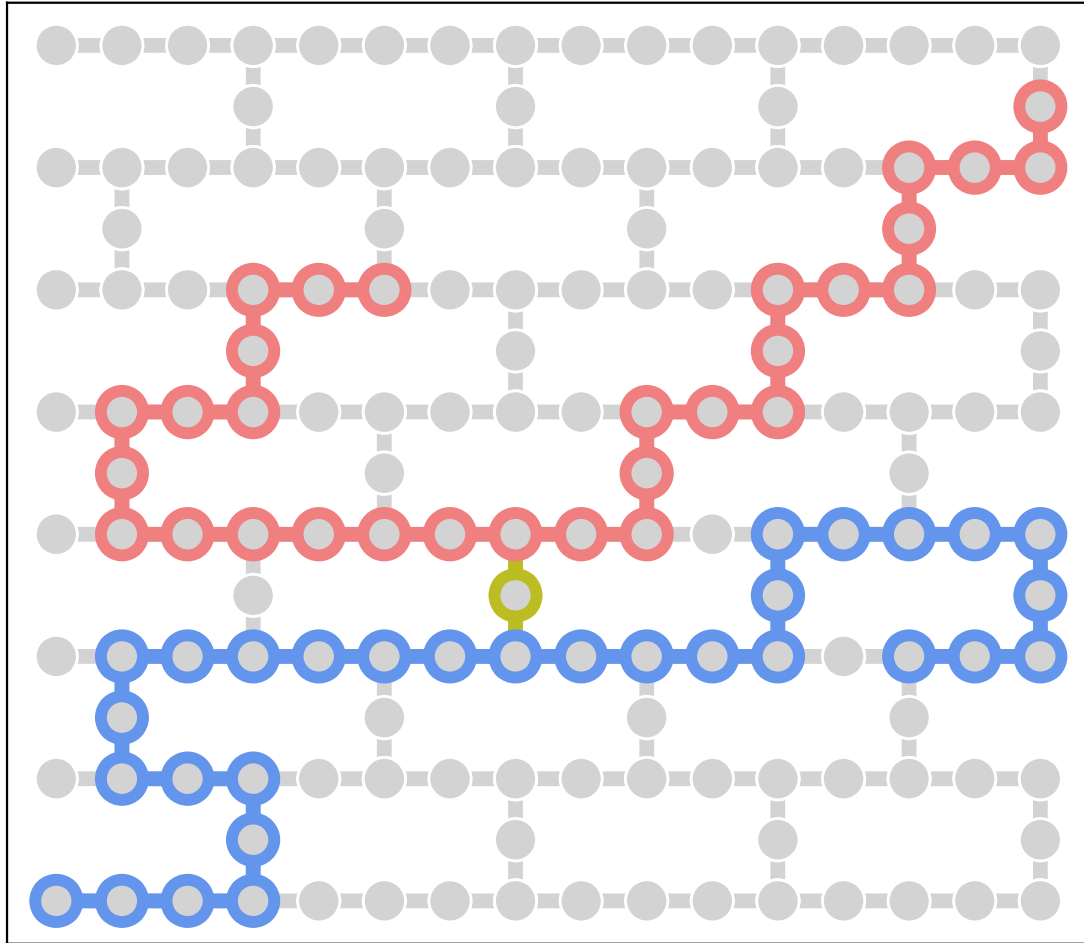
- Action of fermionic Gaussian unitaries (orbital rotations) applied efficiently classically

$$\hat{H}(\kappa) = e^{\kappa_{pq} \hat{c}_p^\dagger \hat{c}_q} \hat{H} e^{-\kappa_{pq} \hat{c}_p^\dagger \hat{c}_q}$$

- In the absence of hybridization (1-sparse):
  - Rotate to basis that diagonalizes the bath “plane waves”
  - The impurity Hamiltonian remains diagonal
- Natural orbitals may increase sparsity

# First SKQD experiments: impurity models

1 Imp. 41 Bath modes  $\rightarrow$  84 + 1 qubits

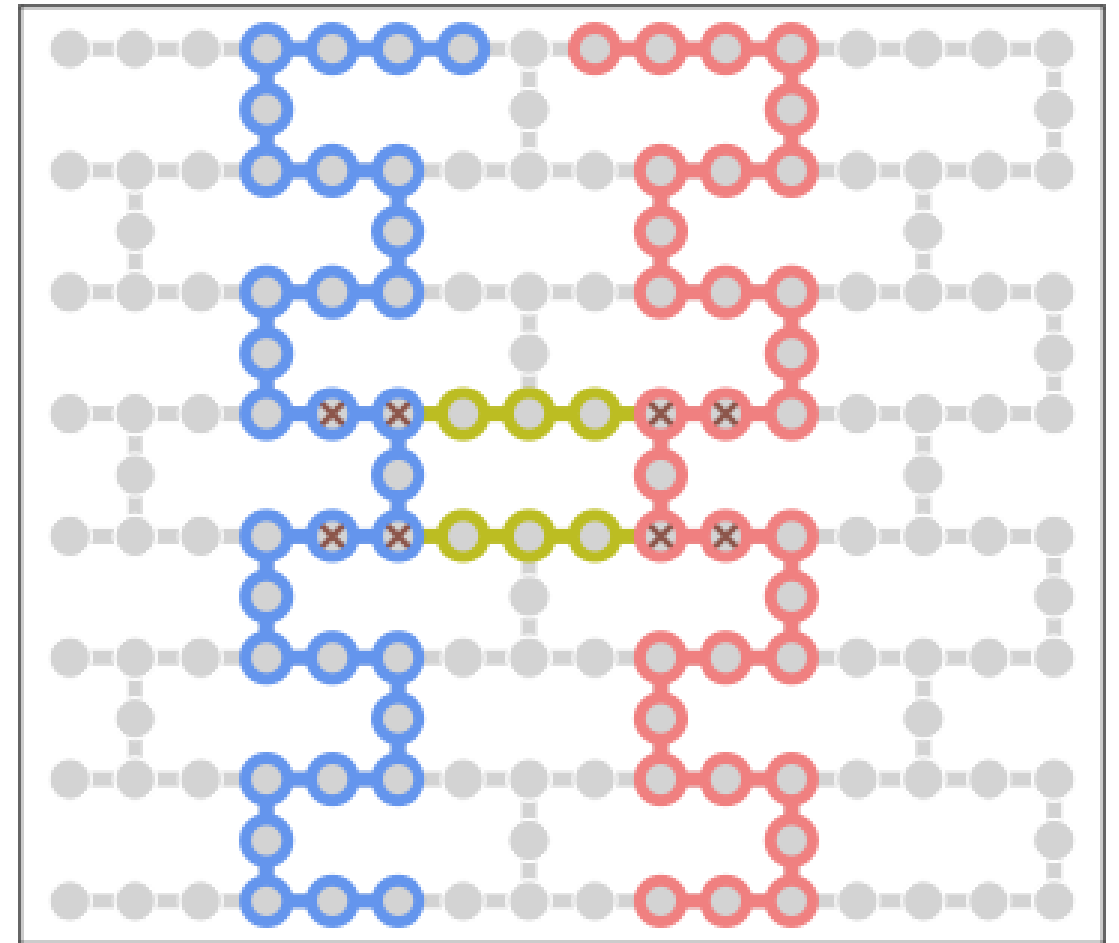


 Spin-up orbitals

 Spin-down orbitals

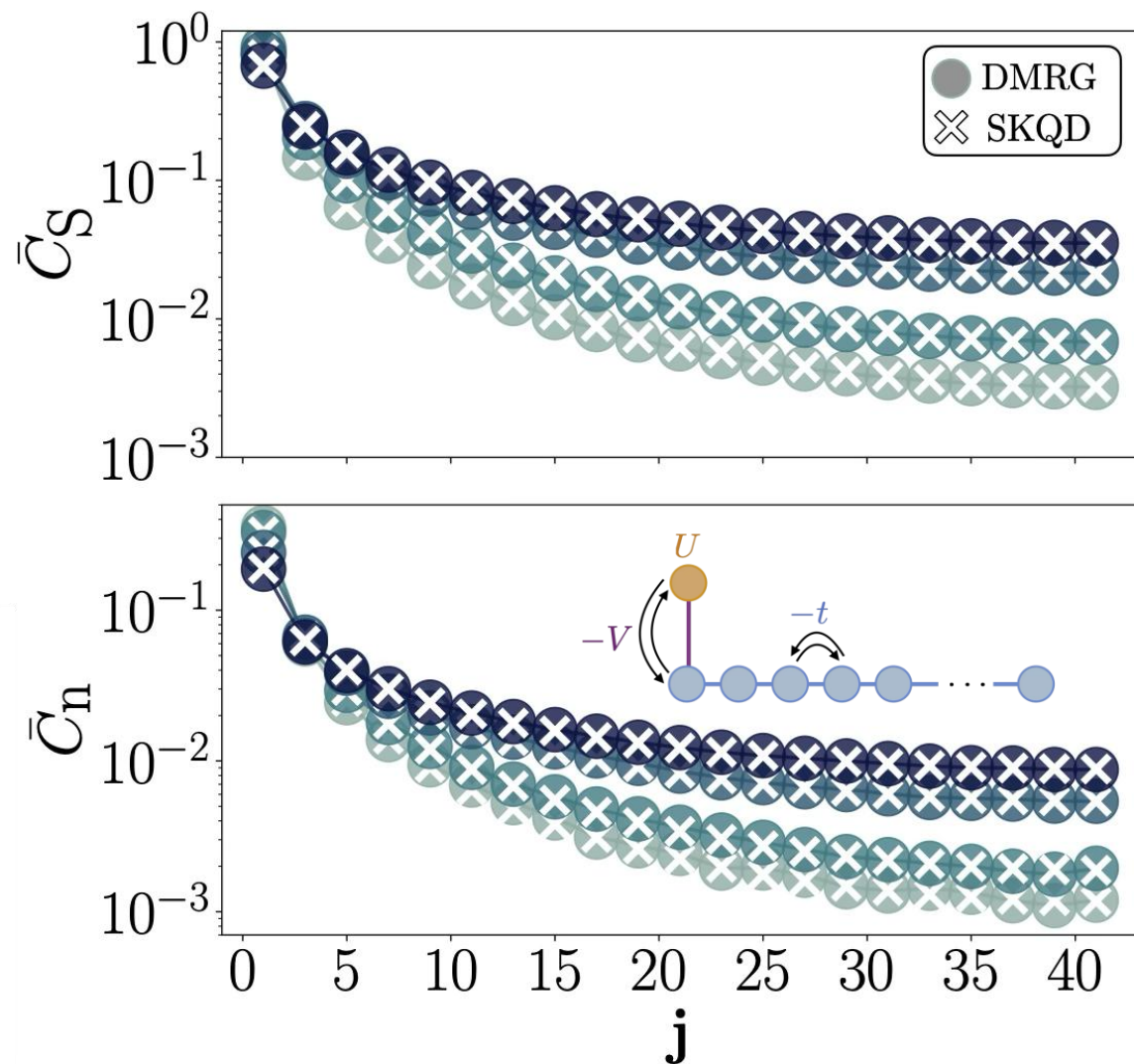
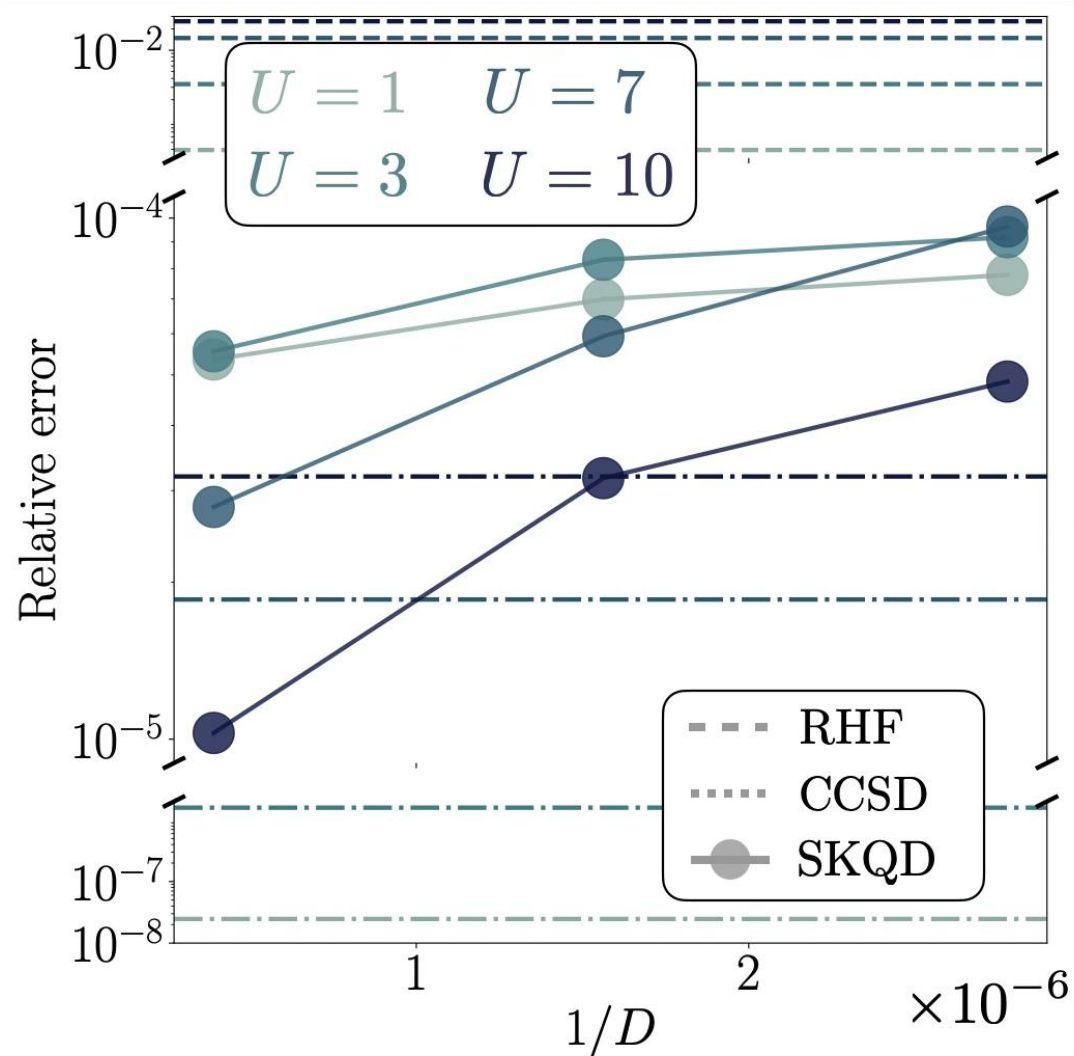
 Ancilla qubits

4 Imp. 28 Bath modes  $\rightarrow$  64 + 6 qubits



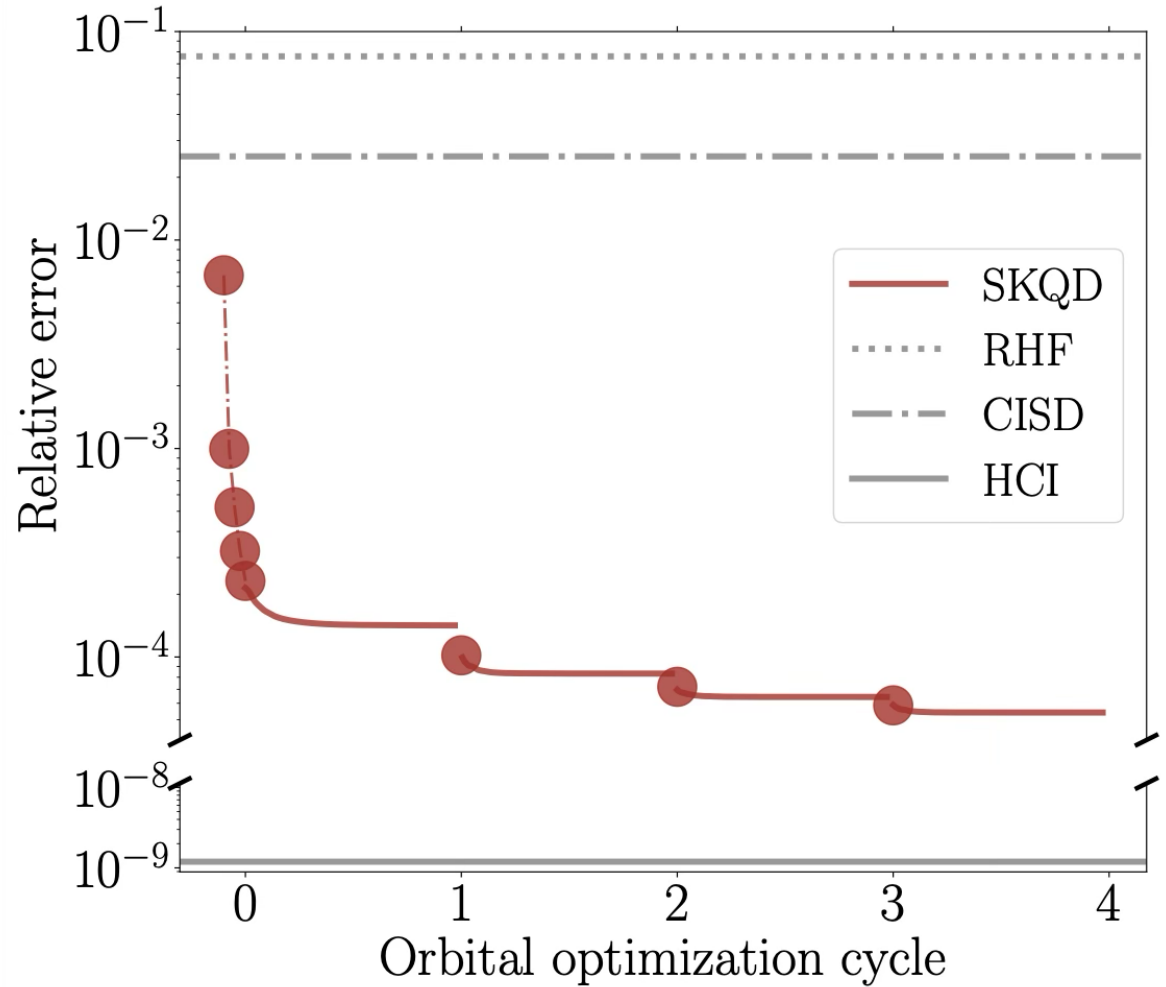
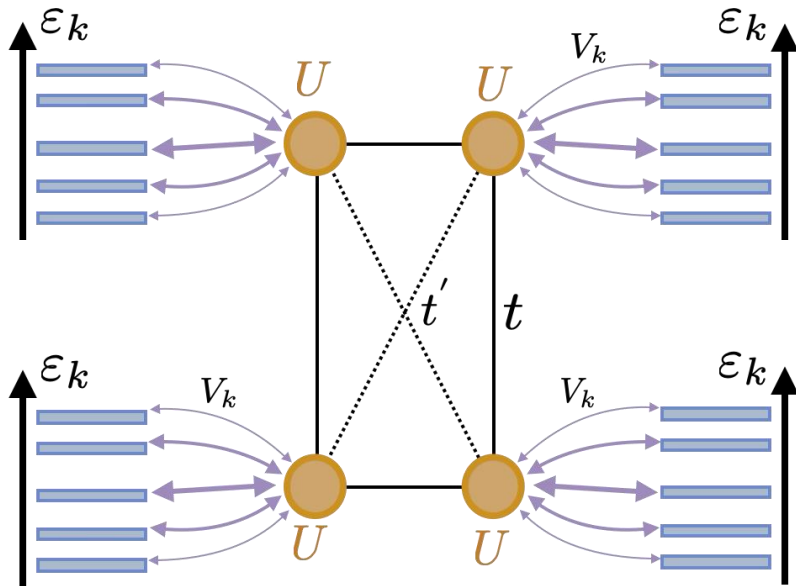
# First SKQD experiments: results

$$\text{Relative error} = \left| \frac{E_{\text{SKQD}} - E_{\text{DMRG}}}{E_{\text{DMRG}}} \right|$$



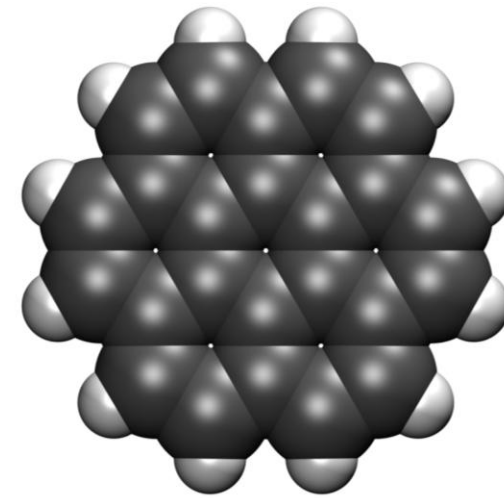
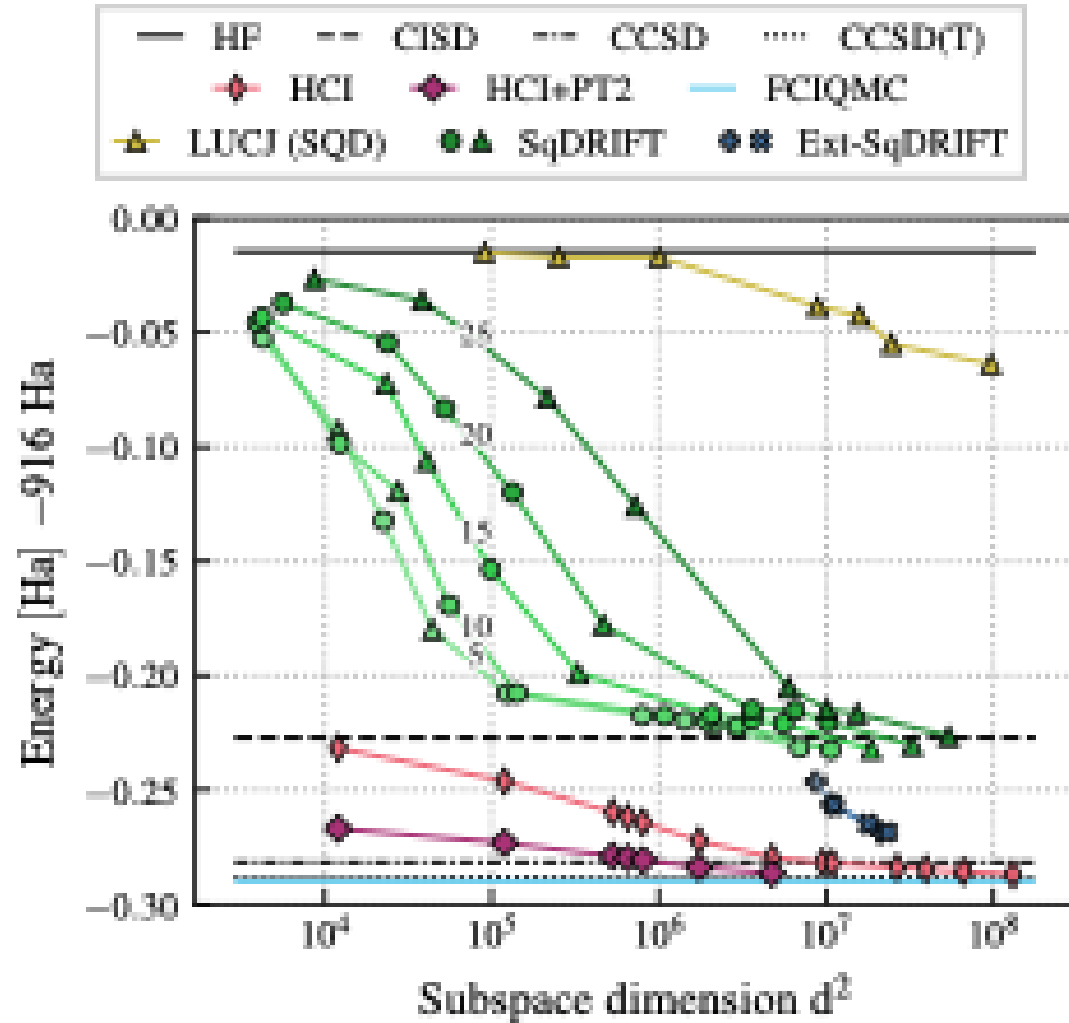
# First SKQD experiments: results

$$\text{Relative error} = \left| \frac{E_{\text{SKQD}} - E_{\text{DMRG}}}{E_{\text{DMRG}}} \right|$$



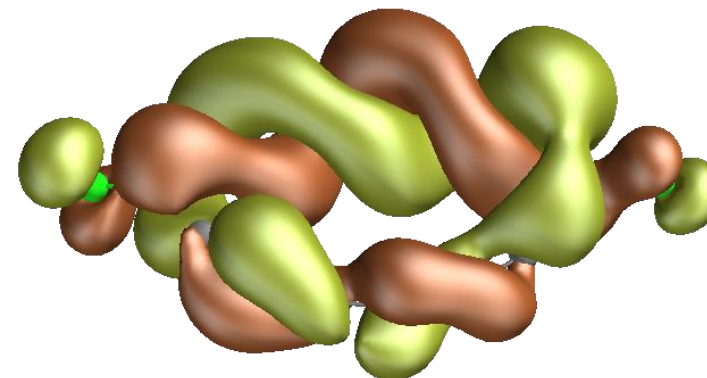
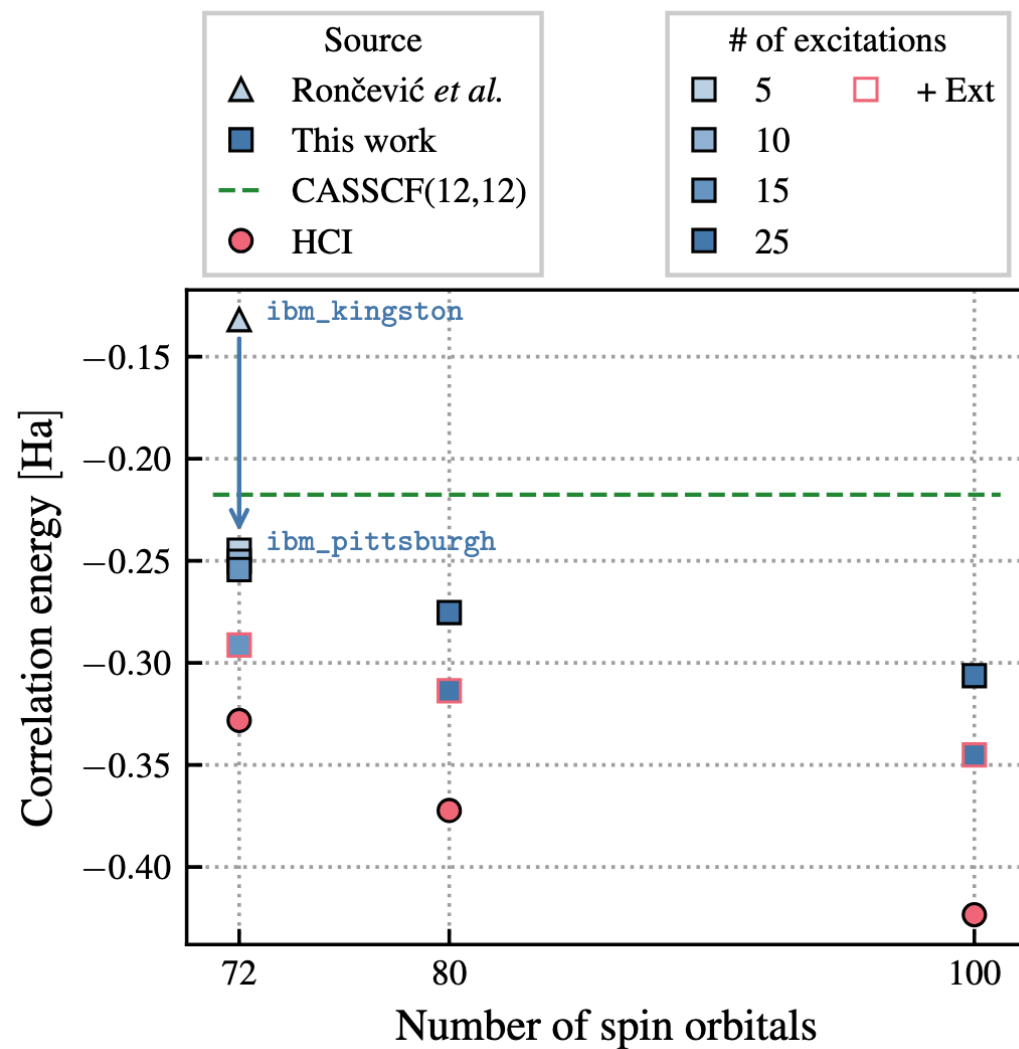


# SqDRIFT experimental results



Coronene active space of  
pi orbitals: 24 electrons  
in 24 orbitals

# SqDRIFT experimental results



Dyson orbital  
computed with  
SqDRIFT on 100  
qubits

# Code and Tools



Qiskit addon: sample-based quantum diagonalization (SQD)  
0.12.0

Search

## Qiskit addon: sample-based quantum diagonalization (SQD)



[Qiskit addons](#) are a collection of modular tools for building utility-scale workloads powered by Qiskit.

This package contains the Qiskit addon for sample-based quantum diagonalization (SQD) – a technique for finding eigenvalues and eigenvectors of quantum operators, such as a quantum system Hamiltonian, using quantum and distributed classical computing together [1-5]. This technique can be run on current quantum computers and has been shown to scale to problem sizes beyond what was possible with variational methods and even beyond the reach of exact classical diagonalization methods [1,2].

# Tutorials



Qiskit addon: sample-based quantum diagonalization (SQD)  
0.10.0

Search

## Improving energy estimation of a chemistry Hamiltonian with SQD



In this tutorial we implement a [Qiskit pattern](#) showing how to post-process noisy quantum samples to find an approximation to the ground state of a chemistry Hamiltonian: the  $N_2$  molecule at equilibrium in the 6-31G basis set. We will follow a [sample-based quantum diagonalization approach](#) to process samples taken from a 36-qubit quantum circuit ansatz (in this case, an LUCJ circuit). In order to account for the effect of quantum noise, the configuration recovery technique is used.

The pattern can be described in four steps:



Qiskit addon: sample-based quantum diagonalization (SQD)  
0.10.0

Search

## Improving energy estimation of a Fermionic lattice model with SQD ¶



In this tutorial we implement a [Qiskit pattern](#) showing how to post-process noisy quantum samples to find an approximation to the ground state of a Fermionic lattice Hamiltonian known as the single-impurity Anderson model. We will follow a sample-based quantum diagonalization approach to process samples taken from a set of 16-qubit Krylov basis states over increasing time intervals. These states are realized on the quantum device using Trotterization of the time evolution. In order to account for the effect of quantum noise, the configuration recovery technique is used. Assuming a good initial state and sparsity of the ground state, [this approach is proven to converge efficiently](#).